

Exercise #2

Introduction To Artificial Intelligence

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For these problems, we split the river into two sections into two section: Left and Right. Left is assumed to be the initial state.

1 Problem Formulation

Problem Statement. *Precisely formulate the problem.*

1. A method of describing a state would be with the tuple: $(C_L, M_L, B_{L/R})$, where:

C_L = The number of Cannibals on the Left

M_L = The number of Missionaries on the Left

$B_{L/R}$ = The location (Left/Right) of the boat

However, we restrict these numbers to as follows:

We do not care about the number of Cannibals/Missionaries on the right, because we can calculate them as follows:

$$C_R = 3 - C_L \quad M_R = 3 - M_L$$

Although these values are not strictly within the state description, from here forth (for readability purposes) we act as if it is, mentally replacing C_R and M_R with these values. The final description is still $(C_L, M_L, B_{L/R})$, and these are merely computed properties of the state.

Note, however, because we want to restrict our state space to only valid states, we place the following restrictions:

$$M_L \geq C_L \quad \text{and} \quad M_R \geq C_R$$

Furthermore,

$$B_{L/R} = \text{Left} \iff M_L + B_L = 6 \quad B_{L/R} = \text{Right} \iff M_R + B_R = 6$$

2. The initial state is as follows:

$$(C_L, M_L, B_{L/R}) = (3, 3, L)$$

3. Our ACTIONS are defined as follow. We use synonyms $L \leftrightarrow \text{Left}$ and $R \leftrightarrow \text{Right}$.

Basically, we check to see if we can send one of both **Cannibal** and **Missionary**. If we can, add them to the solution set. Next, we check to see if we can send either 1 **Cannibal** and 1 **Missionary**. After, we check to see if we can send 2 **Cannibal** and 2 **Missionary**. Because there are rules we have to account for (i.e., there must be more **Missionary** than **Cannibal** *unless there's 0 Missionary*).

```

1: function  $\neg(\text{direction})$ 
2:   if direction = Left then
3:     return Right
4:   else
5:     return Left
6:   end if
7: end function
8:
9:
10: function ACTIONS( $s$ )
11:   PossibleActions  $\leftarrow \{\}$  ▷ A set of sets of moves
12:
13:   for all direction  $d \in \{\text{Left}, \text{Right}\}$  do
14:     if  $a.M_d > 0$  and  $a.C_d > 0$  then
15:       NewAction  $\leftarrow \{\text{send } \text{Cannibal } d \rightarrow \neg d\}$ 
16:       NewAction  $\leftarrow \{\text{send } \text{Missionary } d \rightarrow \neg d\}$  ▷ Note this has the form
17:        $\{\dots, \dots\}$ 
18:       PossibleActions  $\leftarrow \text{PossibleActions} \cup \{\text{NewAction}\}$  ▷ Note this has the
19:       end if form  $\{\dots, \{\dots, \dots\}, \dots\}$ 
20:
21:   for  $i \leftarrow 1$  to 2 do
22:     if  $a.M_d - i \geq 0$  then ▷ Is there even enough missionaries to remove?
23:       if  $(a.M_d - i \geq a.C_d$  or  $a.M_d - i = 0)$  and  $a.M_{\neg d} + i \geq a.C_{\neg d}$  then
24:         NewAction  $\leftarrow \{\text{send } i \text{ Missionary } d \rightarrow \neg d\}$ 1
25:         PossibleActions  $\leftarrow \text{PossibleActions} \cup \{\text{NewAction}\}$ 
26:       end if
27:     end if

```

¹Note by saying “send i **Missionary**”, we are just saying “send a missionary” i times. For $i = 2$, we have NewAction $\leftarrow \{\text{send a missionary}, \text{send a missionary}\}$.

```

28:
29:         if  $a.C_d - i \geq 0$  then           ▷ Is there even enough cannibals to remove?
30:             if  $a.C_{\neg d} + i \leq a.M_{\neg d}$  or  $a.M_{\neg d} = 0$  then
31:                 NewAction  $\leftarrow \{\text{send } i \text{ Cannibal } d \rightarrow \neg d\}$ 2
32:                 PossibleActions  $\leftarrow$  PossibleActions  $\cup \{\text{NewAction}\}$ 
33:             end if
34:         end if
35:     end for
36: end for
37:
38:     return PossibleActions
39: end function

```

4. We define $\text{RESULTS}(s, a)$ as follows.

```

1: function RESULTS( $s, a$ )
2:      $a' \leftarrow a$ 
3:      $a'.B_{L/R} = \neg a'.B_{L/R}$ 
4:
5:     for all action  $\in a$  do           ▷ We defined an action as a set of steps
6:         if action = send Cannibal Left  $\rightarrow$  Right then
7:              $a'.C_L = a'.C_L - 1$ 
8:         end if
9:
10:        if action = send Cannibal Right  $\rightarrow$  Left then
11:             $a'.C_L = a'.C_L + 1$ 
12:        end if
13:
14:        if action = send Missionary Left  $\rightarrow$  Right then
15:             $a'.M_L = a'.M_L - 1$ 
16:        end if
17:
18:        if action = send Missionary Right  $\rightarrow$  Left then
19:             $a'.M_L = a'.M_L + 1$ 
20:        end if
21:    end for
22:
23:    return  $a'$ 
24: end function

```

5. We define $\text{ISGOAL}(s)$ as follows.

```

1: function ISGOAL( $s$ )

```

²See previous footnote.

```
2:   return  $s = (0, 0, \text{Right})$ 
3: end function
```

6. We define $C(s, a, s')$

```
1: function  $C(s, a, s')$ 
2:   return  $|a|^3$ 
3: end function
```

2 State Space

Figure 1 provides the complete space for the problem, with **M** representing Missionaries and **C** representing cannibals.

3 Possible Solution

Figure 2 outlines a path for a solution, with **M** representing Missionaries and **C** representing cannibals.

³We define the $|x|$ operator to return the length of x .

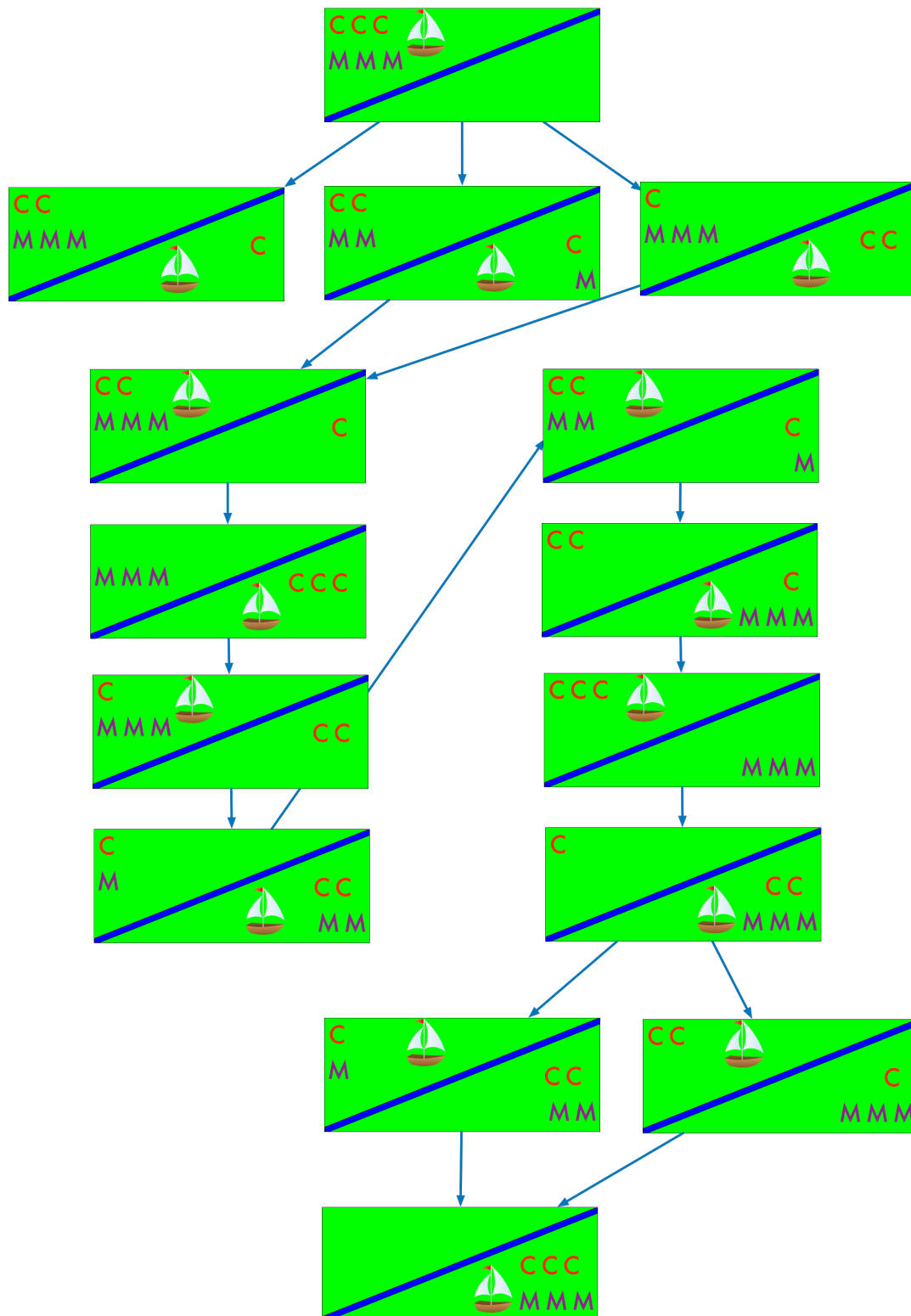


Figure 1: The Complete State Space

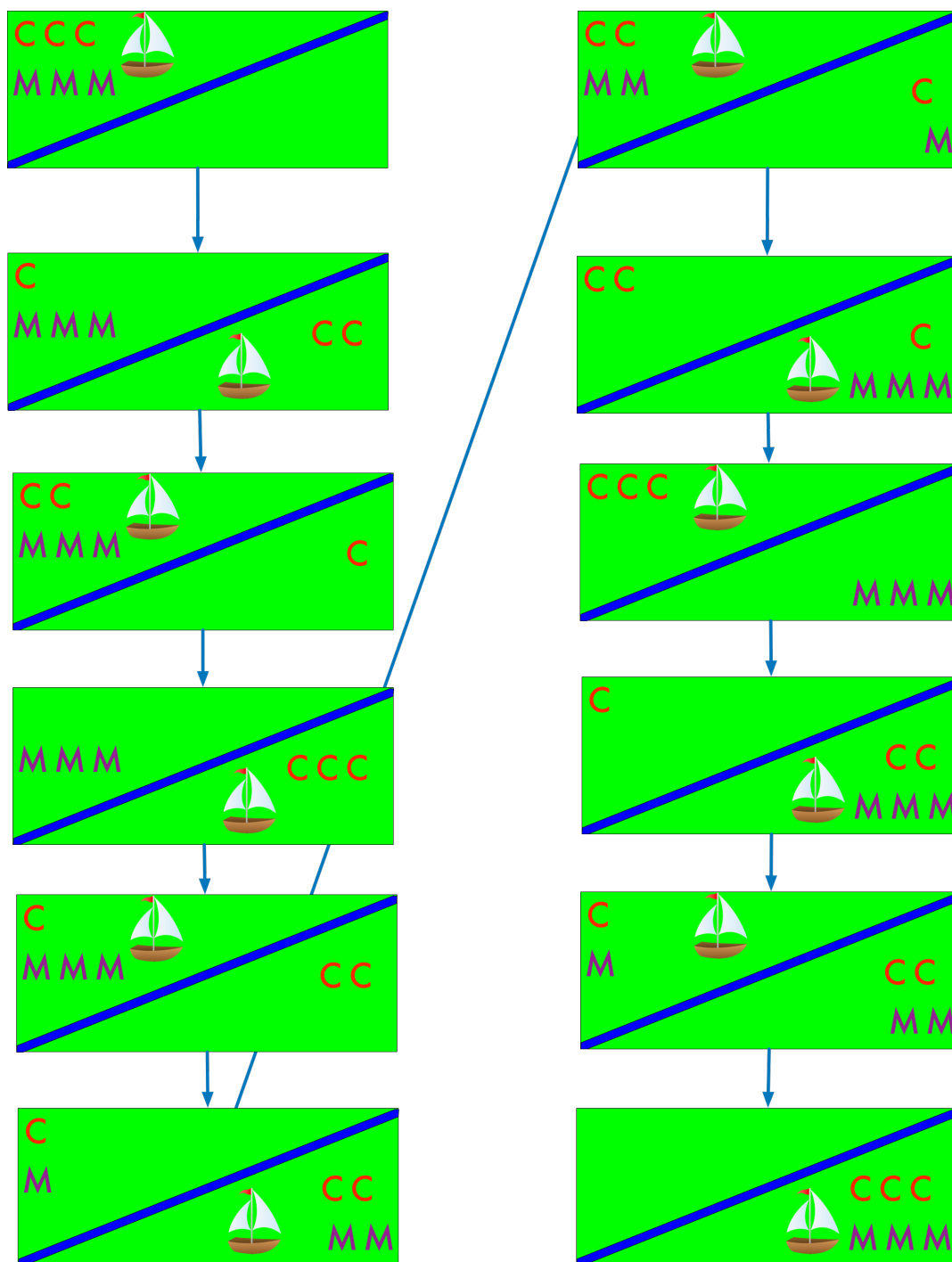


Figure 2: Path To A Solution